Chapter 3

Leveraging AI in Education

IN THIS CHAPTER

- **Adapting work and school approaches to AI use**
- **Understanding critical thinking strategies**
- **Creating personalized learning opportunities**
- **Exploring ideas for educators**
- **Grasping why banning AI from schools is a bad idea**

When generative AI (GenAI) was introduced, students immediately began using it to do their homework. However, many chose to use it not to *help* with their homework but to do it *for* them. Educators weren't amused. A kerfuffle followed as educators and parents fretted that AI was an instrument for cheating and would rob students of their ability to hone critical thinking skills. The worriers were only partly right.

Cheaters are going to cheat, and teachers need to catch them if they can.

O'REILLY

text that's difficult to distinguish from human writing. However, educators are adept at catching cheaters despite their ever-changing methods. For decades, educators and teaching assistants have confiscated crib notes, barred calculator use in classes and tests, busted test ringers, checked dutifully for plagiarism, and generally outed the ne'er-do-wells

and the ethically impaired. So, really, GenAI brings nothing new to the ongoing need to check for cheaters.

I urge educators and employers to focus on grading or evaluating work based on accuracy, originality, execution, and quality instead of trying to figure out whether an AI tool was used in the process.

Therefore, in this chapter, I don't spend time covering how to cheat because that will be sorted in due time. Instead, this chapter is about how educators can use GenAI to better serve their students by offering more personalized support and instruction. It's also about how GenAI can reinvent education. That's probably why (without assuming, of course) you are reading this book — to get the most out of emerging technology. The prompting techniques covered in Book 2 will help you use GenAI in both how and what you teach in the age of AI. You might also want to take a look at Book 3, Chapter 5, to see how different disciplines use GenAI to further help you understand how your students may need to use tools like ChatGPT after graduation.

Using AI Is Here to Stay

Regarding the use of AI for completing assignments, the harm is more likely to come from the use of AI detection tools in the chase for cheaters rather than from cheaters slipping a machine-generated assignment past the gatekeepers. To date, I've never seen an AI detection tool that's accurate or even consistent in labeling works as either AI- or human-made. Yet far too many educators and employers treat these tools as infallible and wield the results as the final word.

Avoiding false accusations of cheating



WARNING Unfortunately, the people who are falsely accused by these AI detection tools typically have no recourse and can never prove their innocence. These unfortunate souls find their careers and school records ir-

revocably stained. The harm done by a false positive result is almost incalculable. This practice needs to come to an immediate halt.

To be clear, yes, GenAI does repeat phrases, words, and sentence structures. But so do people. There are no set writing patterns, word choices, sentence structures, or anything else that's uniquely and undeniably associated with GenAI outputs. That's why AI detection tools don't work. There's no sure "tell" to find.

Spotting AI in use by students is a worthless exercise anyway. It's now part of today's existence, for better or worse. Using AI effectively is now a core mission for educators to teach students and a fundamental tool all students must master if they're to navigate their daily lives or be employed.

Accepting the role of AI at school and work

Make no mistake, both educators and students will outlive any current generative AI tool. These early manifestations continue to morph into something more than the sum of their new features or are replaced by competing tools, such is the nature of all living and invented things. Even so, AI is reshaping the world and is firmly entrenched as an integral part of it. Whether AI's presence today is more insidious or illuminating depends entirely on how you see it and use it. For a balance to be struck or the scales to be tipped in favor of all that is good, educators must step up and teach that. You can't teach that if you keep attacking AI use in schools like it's evil incarnate.

AI will be a permanent fixture in life going forward. Of course, it may fade into the background and become the backbone of all the tools rather than function as a standalone as GenAI tools currently do. But AI will never go away. As the Borg on *Star Trek* would say, "Resistance is futile." Better to embrace it and make rules assuming that AI will be used than waste your time and your students' academic lives trying to forbid it and root it out.



REMEMBER GenAI use requires skill in to get skill out. Yes, that's my way of saying that "garbage in, garbage out" still applies. The quality of the work speaks directly to the skill level of the person who made it, regardless of the type of tool they used. If the quality of work, the originality of thought, the uniqueness in its creativity, or a flawless execution of the task is present, then the person who made it — with or without GenAI's help — earned a grade to match.

Changing the Structure of Education

Speaking of upcoming changes in educational institutions and processes: Some major disruptions are easy to foresee. But these changes won't be the last. Ripple effects will spread across decades as both AI and educators evolve to meet the challenges and opportunities of the day.

GenAI's integration into education is poised to transform both institutional structures and teaching methods significantly. At the core of this transformation is the shift toward personalized learning. Traditionally, students progress through a standardized curriculum at the same pace, but AI allows learning to be tailored to individual needs, pacing, and level of understanding. GenAI can act as a personalized tutor, responding to each student's questions, identifying areas for improvement, and adapting content dynamically. Over time, this can lead to a shift away from age- or grade-based progression toward competency-based advancement, in which students move forward upon mastering the material, which isn't based on time spent in a classroom or memorization skills.

Using AI in tutoring, testing, and updating curriculum

Tutoring and support are also likely to change drastically. Many students currently rely on scheduled, often costly tutoring sessions, which may not always be accessible. GenAI is constantly available as long as limits in tokens or chats aren't exceeded. Provided schools make GenAI accessible to all, even students with little to no private resources can receive instant support, explanations, and feedback. This shift might lead institutions to

reconsider how they allocate all resources, potentially allowing teachers to focus more on specialized lessons and support while GenAI handles foundational content review and reinforcement outside the classroom.

Traditional assessment tools might undergo an even more dramatic change. Today's reliance on standardized tests means that knowledge is evaluated periodically, which may not always reflect how much information students understand, retain, and can successfully apply now or going forward. GenAI, however, can offer immediate and ongoing feedback on a student's performance. It can also present the same information in different ways to match individual learning styles to help students grasp difficult concepts and overcome learning obstacles.



REMEMBER If you're an educator, instead of high-stakes testing, you can move toward continuous assessment, in which students receive ongoing evaluations that may better capture their learning progress. This type of assessment can significantly reduce test-related stress and focus more on actual comprehension and critical thinking development.

The traditional curriculum, which often lags behind the latest knowledge and trends in the real world, also stands to benefit from AI. GenAI's access to a vast amount of information allows you to easily update lesson plans and incorporate the latest research. AI can help ensure that students learn content relevant to their futures, making curriculums more dynamic. This potential for fluid content creation allows you and other teachers to focus on more interactive and practical applications rather than spending time updating static content. Such an approach is better and more efficient in helping students stay aligned with current developments.

Preparing for shifts in educator roles

Your role as an educator will likely shift along with approaches to education. You'll act less as a content deliverer and more as a mentor and learning facilitator. As AI tools take over foundational teaching tasks,

you'll have more time to focus on guiding critical thinking, honing intuitive intelligence, sharpening research and fact-checking skills, promoting socio-emotional learning, and cultivating creativity. This role evolution may lead to changes in hiring and training practices, with a growing emphasis on interpersonal skills and adaptability, as well as mastery in integrating AI into classroom management and lesson plans.

Gaining efficiency with AI help for administrative tasks

Administrative tasks, which currently consume significant resources in education, will also likely see gains in efficiency with AI. To be clear, GenAI tools generally can't automate anything beyond the text and images they generate. For example, ChatGPT, at its core, is a text generator, but it can manage some image work behind the scenes with the aid of other GenAI tools, such as the sister application DALL-E. I point that out only to make you aware that you'll need to use GenAI outputs in other software to fully leverage its capabilities.

For example, GenAI can help you with grading student work and tests, tracking student progress, and managing communications with students and parents. It can't complete the tasks for you, but by using GenAI alongside other automation apps, or embedded in specialized automation apps, or as the UI and orchestrator of autonomous AI agent systems, you can automate these processes. Never fear: GenAI is already embedded in most work software, so you have plenty of options available. More will appear on the market soon.

Any way you choose to use GenAI to lighten the administrative load will be instrumental in reallocating resources toward enriching educational experiences or reducing operational costs. This, coupled with AI-enabled flexibility, may give rise to more hybrid learning models, allowing students to participate in a blend of digital and physical classrooms with digital and physical instructors present in either or both environments.

Providing equal access to quality instruction

By making quality education more accessible to geographically isolated or underserved students, AI can contribute to a truly globalized educa-

tion system. GenAI can act as an equalizer, providing access to quality resources and instruction regardless of location, which can lead to an expansion of international, online educational platforms and credentialing systems. Today, when several countries are experiencing aging populations and too few young people to fill jobs and drive each country's progress further, global education may go far in tapping into highly skilled and educated labor forces in countries with younger populations. Conversely, it can help older people teach and store skills and information that may otherwise be lost to time.

As AI becomes more integrated into education, institutions may transform into adaptable, personalized, and globally accessible models. Instead of rigid structures, schools may become more fluid environments that equip students with skills for an increasingly dynamic world, reshaping both the experience and the objectives of education.

Flipping the Teaching Model

As I mention in the chapter introduction, some educators, parents, and employers are concerned that students who use AI will miss out on developing critical thinking skills. This problem will likely self-correct to a degree, but it can also be prevented by flipping the teaching model.

Currently, students are taught information and then quizzed. With GenAI, the student must first ask the right question to unlock the information, an approach that reverses how critical thinking is traditionally taught and tested.

GenAI's outputs are only as good as the inputs that a teacher or a student gives the tool. To prompt the AI to deliver the answers they seek, students need to think carefully and critically about how to word their query or command. Teaching them how to think about and write prompts is an effective albeit different way to help them develop critical thinking skills.

Consider and compare the following two examples:

1. *Basic prompt:* "Generate an image of a cat."

This prompt is straightforward, but it lacks details. The outcome could be any cat — small, large, fluffy, hairless, sitting, jumping, indoors, or outdoors. The result may not be specific to what you want.

2. *Better, strategic prompt:* "Generate an image of a fluffy, orange tabby cat with green eyes sitting on a windowsill. The cat should be looking out at a rainy cityscape in the background, with drops of rain visible on the glass. The setting is cozy, and the cat is surrounded by indoor plants and books."
   Version 2 is much more detailed and specific. By specifying the cat's appearance, the setting, and even the mood (cozy), you increase your chances of getting an image that matches your vision. This prompt demonstrates critical thinking by considering factors like context, environment, and cat's characteristics.

The strategic prompt in the second example illustrates why it's important to think critically: The more detail and clarity you provide, the more likely you are to receive output that meets your expectations. It's about guiding the outcome and ensuring the final product aligns with your vision.

This comparison illustrates why critical thinking helps you get the most out of GenAI. You really can't get far by being vague or lazy about prompting. That's why I advocate grading the quality of a student's work over whether they used tools like Word or ChatGPT. The outcome is invariably the result of their own effort with or without AI tools.

And so, you can teach critical thinking, at least in part, by teaching prompting skills. Here are some benefits of this approach:

- **It's a win-win for students** because both critical thinking and prompting skills are in high demand by employers. These skills can also improve the quality of life by elevating students' performance in daily life problem-solving.
- **Prompting and critical thinking will remain necessary and in high demand** for the lifetime of students and likely for generations afterward. Although writing prompts will soon be automated by AI, too, critical thinking will remain as important in commanding and overseeing AI agents as it is now in writing prompts. Consider this scenario:
  1. AI like AutoGPT can write prompts, which are useful down the chain of AI agent systems wherein one AI orchestrates several specialized AI bots to work together to complete a bigger task.
  2. A user must initiate the action and approve AI decisions along the way. In other words, you would have to write a prompt to get the party started, and then that AI could automate prompts to invite, orchestrate, and manage other AI agents to join in.

3. A human will still have to approve or correct AI agents' actions along the way.



REMEMBER The evolution of AI agent systems will go a long way toward making AI benefits accessible to everyone, but autonomous AI agents won't eliminate the need for creative and critical thinking. In this age of AI, those who can wield these tools well will prosper, which makes critical thinking skills and prompt engineering excellent job skills for educators to teach — and for educators and students to learn.

Leveraging GenAI to Aid Overworked Educators

If you're an educator, you're notoriously overworked and underappreciated, often juggling a multitude of responsibilities that extend far beyond the classroom and the workday. The advent of GenAI AI tools such as ChatGPT offers a promising solution to alleviate some of this strain.

Getting help with grading assignments

GenAI can assist in grading assignments with a level of depth and analysis that you may not have the time to provide. Further, AI can deliver test scores and evaluations of other forms of student work along with detailed feedback for you and students to act on.

But this process isn't nearly as arduous as you're probably imagining. Indeed, teachers have been quietly using GenAI to grade tests and papers ever since it came out. But it's only recently that schools are beginning to encourage its use. Here's one way you can see for yourself how this process might work for you in grading papers if you were to use ChatGPT Plus to do it. (You can't grade multiple papers in a single prompt, so taking these manual steps may be too slow if you have a large class and many papers to grade.)

1. **Prepare the paper to grade for review by ChatGPT Plus.**
   Depending on whether student "papers" (that is, homework or thesis papers or whatever it is that you're grading) are in digital form or written on an actual

sheet of paper, you can

1. Take a screenshot or a picture of the paper.
2. Upload the digital document.



2. REMEMBER **Label and save the image or the document.**
3. **Attach the saved image to the prompt bar.**
4. **Type instruction such as "Grade this paper" or "List correct and incorrect responses on the attached image."**
   If you prompt for a list of right and wrong answers, you can calculate the grade yourself.
5. **Review and fact-check the response.**
   Essentially, make sure that you agree with the score.
6. **Repeat the process on the other student papers.**

Of course, more tools are appearing on the market every day, offering efficient ways to use GenAI for grading tasks. One example is Writable, which can assist you with grading any assignment in Word or Google Doc formats. It comes with more than a thousand prompts, assignments, and the Houghton Mifflin Harcourt curriculum built in for your use. You can edit them, too, if you want. Essaygrader.ai is another GenAI tool specializing in, you guessed it, grading essays.

I'm not endorsing any particular GenAI tool. I'm just trying to give you a sense of what's available. Keep an eye out because more options will hit the market over time.

Exploiting AI for efficient mentoring

GenAI can readily access and identify areas in which each student is struggling and offer personalized suggestions for improvement. This enables you to develop and execute tailored teaching or mentoring plans for each student, enhancing the overall learning experience and hopefully doing so without adding anything to your already full workload.

Moreover, GenAI can perform these tasks in a matter of minutes. It can compute homework, tests, and project scores quickly. This efficiency means that you and your assistants can use AI to score assignments be-

fore the end of the workday, freeing up valuable time for other important tasks or for personal pursuits in off-duty hours.

Adapting lessons and admin tasks

GenAI is also a valuable tool for developing or adapting lesson plans. You can generate comprehensive and detailed lesson plans on-demand and fit them to your own requirements as the instructor or to the needs of an individual student. Further, you can customize lesson plans to fit specific classroom needs or a change in learning environments. For example, if you're teaching a unit on the American Revolution, a GenAI tool such as ChatGPT-4o can provide a structured lesson plan complete with objectives, activities, video or audio avatar reenactment scripts, and assessment methods. If you prefer a more concise version, ChatGPT can condense the information accordingly.

Beyond classroom instruction, GenAI can assist you with various administrative tasks. From making a list of classroom supplies needed for specific lessons to completing paperwork required by school administrators, GenAI streamlines these processes, allowing you to focus more on teaching and less on bureaucratic tasks.



TIP As an educator, you can use AI efficiently to get your private time back without compromising the quality of education you deliver.

Changing How Subjects Are Taught

GenAI is significantly transforming education now and for the future. This transformation is visible in several key areas that involve adapting learning content, including

- **Personalized learning:** GenAI can adapt educational content to meet the needs of individual students, providing tailored explanations, exercises, and feedback based on each student's level of understanding. For example, a student struggling with algebra might receive step-by-step explanations and additional practice problems specifically designed to address their weak points.

- **Learning style and access:** GenAI can adapt content according to educational standards set forth in VAK (Visual, Auditory, Kinesthetic) and VARK (which adds Reading/Writing) styles. Because GenAI is available 24/7, students can access help at any time, extending learning opportunities beyond traditional classroom hours.
- **Student engagement:** GenAI enhances the learning experience by creating interactive content as well as a variety of gamification plans. Imagine a history lesson in which students can engage in a simulated conversation with a historical figure, asking questions and receiving detailed, context-rich responses. This kind of interaction makes learning more dynamic and engaging. Additionally, students benefit from instant feedback on their work, helping them quickly understand and correct mistakes and accelerating the learning process.



REMEMBER GenAI can also assist in creating worksheets and other educational materials, which saves time and money and typically improves the quality of resources available. For instance, if you're preparing a unit on environmental science, you might use GenAI to generate quizzes and discussion prompts tailored to the specific topics covered in the unit or your field trip.

Providing safer spaces for learning

Language learning is another area in which GenAI is making a substantial impact. Students can practice conversational skills in a new language by engaging in dialogues with the AI, which provides a safe and supportive environment for practice. For example, a student learning Spanish can converse with GenAI, practicing their speaking and listening skills without the fear of judgment. The AI can also explain complex grammar rules and translate phrases, making it easier for students to grasp new languages.

The potential of GenAI is even more expansive. The development of highly sophisticated AI tutors capable of understanding and adapting to a student's unique learning style, pace, and preferences is a certainty on the near horizon. For example, an AI tutor might recognize that a student learns best through visual aids and adjust its teaching methods accordingly, using more diagrams and visual explanations.

Immersive learning environments can also become commonplace, with GenAI integrated into virtual reality (VR) and augmented reality (AR) experiences. Imagine a biology class in which students can virtually explore the human body in 3D or a history lesson in which they can walk through ancient Rome. These immersive experiences make learning more engaging and memorable. Additionally, AI-driven gamified learning platforms can incorporate elements of game design into the curriculum, motivating students through challenges and rewards.

Inclusive education stands to benefit greatly from AI advancements, too. GenAI can provide support for students with disabilities or whose learning journey was interrupted for one reason or another, causing them to fall behind. By offering alternative ways to interact with educational content, these and other obstacles can be overcome. For example, a student with a visual impairment might use text-to-speech features, whereas a student with a hearing impairment might benefit from speech-to-text capabilities.

Further, GenAI can bridge language barriers in diverse classrooms by providing the same lessons in multiple languages as needed, ensuring all students have access to the same quality of education.

Learning collaboratively

Collaborative learning is another area in which AI can make a significant impact. Global classrooms, where students from different parts of the world collaborate on projects, share perspectives, and learn from each other in real time, might become a reality. Imagine a science project in which students from different countries work together to solve a global environmental issue, facilitated by GenAI. GenAI platforms can also support peer-to-peer learning by mediating discussions, providing prompts, and ensuring conversations remain productive and on-topic.

In terms of curriculum improvement, AI can analyze vast amounts of educational data to identify gaps and suggest enhancements. This means that curricula can be continually updated based on new discoveries, societal changes, and technological advancements, keeping education current

and forward-looking. For example, if data shows that students are consistently struggling with a particular concept in physics, educators can be alerted to this issue, and GenAI can provide suggestions, strategies, and lesson plans to address it.



REMEMBER As AI becomes more integrated into education, it's crucial to address ethical considerations such as data privacy, algorithmic bias, and the digital divide. Ensuring that AI tools are used responsibly and equitably is key to maximizing their positive impact on education.

In short, you can leverage GenAI to provide a more effective, enjoyable, and accessible experience for all learners. But that requires that you take a proactive role now in plotting the path forward as an educator. Remember, GenAI does nothing until prompted to do so and then only as prompted. It's up to education leaders and futurists to decide where education is going with AI.

Supporting Special Education Needs

GenAI presents immense potential in special education by helping to tailor support for students with diverse learning needs. Specialized GenAI tools are better for these tasks than the generalized version. It's not as if you can hand any kid a tablet with ChatGPT on it and expect everything to work out somehow. As an educator, you need to design GenAI tools and chatbots to specifically work with special-needs students. You might be able to do that on your own, but you'll probably be more comfortable and successful working with AI scientists or an AI-skilled technologist to make it happen. Look for vendors to produce GenAI tools aimed at delivering services that uniquely assist with this student group.

Here are some major benefits of using GenAI in special education:

- **Its ability to create customizable learning aids.** For instance, students with dyslexia can benefit from personalized reading exercises that adapt to their pace and learning or coping style. GenAI can generate texts with adjustable complexity, ensuring that students are neither overwhelmed nor under-chal-

lenged. Additionally, it can offer real-time assistance, such as breaking down complex sentences or providing synonyms for difficult words, thereby enhancing the student's reading experience and boosting their confidence.

- **Its capacity to provide behavioral and emotional support.** Students with emotional and behavioral disorders often require consistent support to manage their challenges effectively. GenAI can offer a form of virtual counseling or encouragement. Assigning GenAI to such a role can provide students with strategies to cope with anxiety, stress, or frustration.



REMEMBER GenAI can guide a student through deep-breathing exercises or help them develop a step-by-step plan to address a particular issue they're facing. However, GenAI isn't a substitute for counselors or mental health professionals.

- **It's role as a nonjudgmental listener,** which allows students to express their thoughts and feelings freely. This aspect can be particularly therapeutic for those who might find it difficult to open up to others. Just be aware of the serious privacy concerns in sharing private and vulnerable information with any AI tool.

Furthermore, specialized GenAI tools can assist you in supporting students' progress. They can identify patterns indicating that a student is struggling so you can intervene promptly. This proactive approach ensures that students receive the support they need before their challenges become upsetting or more severe.

Delivering Data-Driven Insights for Educators

Data-driven insights have become a cornerstone of enhancing and evaluating teaching effectiveness and student outcomes. GenAI offers one approach to compiling and analyzing data. Indeed, its greatest strength is in data discovery.



REMEMBER However, it's generally a mistake to replace all other analytics and data tools with GenAI. The smarter plan is to add GenAI to your school's tool chest rather than ditch the tools you already use. I say that because there's substantial value in using analytical tools that you're familiar with over one that introduces a rather long learning curve and

sometimes questionable responses. Additionally, the performance, accuracy, and reliability of modern analytics tend to vastly outperform GenAI tools at the moment. There's really no advantage to stopping their use.

Using GenAI tools for efficiency

The role of GenAI tools in education is growing, although their impact on student performance analysis is still under examination. These tools offer several potential benefits. For instance, they can enhance efficiency in task completion, with studies indicating that students, particularly in fields like business informatics, can complete certain tasks more rapidly. Additionally, GenAI can aid students in gathering information and brainstorming ideas more effectively.

However, there are notable limitations and concerns. Research, such as a 2024 study titled "Is it harmful or helpful? Examining the causes and consequences of generative AI usage among university students" published in the *International Journal of Educational Technology in Higher Education*, has linked increased use of GenAI to declining academic performance, as evidenced by lower grade point averages.

Furthermore, the same study found that GenAI usage is associated with increased procrastination and self-reported memory loss. Another significant issue is the need for human oversight. Although GenAI can efficiently analyze text, human judgment remains crucial in interpreting results, particularly in contexts such as course evaluations.

Questionable student progress tracking

Regarding the current state of GenAI in classrooms, the concept of a specialized GenAI tool for comprehensive student progress tracking isn't yet widespread or well established. Although such systems might be in development or undergoing limited testing, they're not commonly used in educational settings. Most research to date focuses on GenAI's impact when used as a general tool rather than as a specialized tracking system. GenAI's ability to accurately track and analyze complex student interactions in real time hasn't been proven or widely implemented.

Additionally, concerns about data privacy, accuracy, and the potential for AI bias in educational settings present significant challenges.



TIP The integration of GenAI tools in education is an evolving field that requires careful consideration of both its benefits and its potential drawbacks.

Banning GenAI Stifles Education

Despite some drawbacks, banning GenAI from schools and education programs is a dire mistake. AI is here to stay, and it's reshaping the nature of work and, indeed, the human experience.



WARNING If you're an educator who doesn't guide students through this transition into effective use of GenAI, you're doing them an injustice. Navigating and functioning in the world without AI skills will soon be as difficult as getting by without computer skills, smartphones, and internet access.

Adopting a nuanced approach to AI

The better course is to acknowledge that change really is the only constant and that it's your duty as an educator to help students master each new turn. Educators are the ones who move a society forward and enable it to adapt. Teach accordingly.

The decision to ban GenAI in educational settings can have far-reaching consequences, stifling innovation, exacerbating educational inequities, and leaving students unprepared for a future where artificial intelligence (AI) plays an increasingly pivotal role.

One of the most significant drawbacks of banning GenAI is the missed opportunities for innovation in education. GenAI has the potential to revolutionize the way students learn by providing personalized and adaptive

learning experiences. For instance, through natural language processing, GenAI can offer tailored feedback on student assignments, helping students understand their mistakes and guiding them toward improvement. This immediate and personalized feedback loop can accelerate learning and comprehension, which traditional methods may not achieve as efficiently.

Additionally, GenAI can assist in creating interactive and engaging educational content, such as virtual simulations and problem-solving scenarios, which can make learning more dynamic and appealing. By banning GenAI, educational institutions may forego these innovative approaches, potentially hindering the evolution of teaching methodologies that better cater to diverse student needs.

Equity and accessibility are also at risk when GenAI is banned. In many educational systems, there's a significant disparity in resources available to students. GenAI can bridge this gap by providing consistent and high-quality educational support to all students, regardless of their socio-economic background. For example, students in underfunded schools may not have access to specialized tutoring or advanced coursework. GenAI can fill this void by offering supplementary educational materials and personalized tutoring sessions, ensuring that all students have the opportunity to succeed. Banning GenAI can widen the educational gap, further entrenching existing inequalities.

Preparing students for the future

Moreover, preparing students for the future workforce necessitates familiarity with AI technologies. As AI becomes increasingly integrated into various industries, the demand for workers who understand and can skillfully interact with AI systems is growing. By incorporating GenAI into the educational curriculum, students gain hands-on experience with AI, developing critical skills that will be valuable in their future careers.

For instance, students can learn how to leverage AI for research, data analysis, and problem-solving, skills that are highly sought after in fields such as technology, healthcare, and finance. Without exposure to AI tech-

nologies like GenAI, students may find themselves at a disadvantage in the job market, lacking the necessary skills to thrive in an AI-driven world.

Bottom line: Banning GenAI in education can have detrimental effects, including missed opportunities for innovation, increased educational inequities, and inadequate preparation for the future workforce. Embracing AI technologies can enhance educational experiences, promote equity, and equip students with the skills they need to succeed in a rapidly evolving world. But GenAI doesn't equal or replace professional human educators. You need both educators and AI to move students to the front line of a quickly evolving and uniquely challenging future.